

NASH EQUILIBRIUM: "JOSH NASH"

A Nash Equilibrium is a situation in a game where each player's strategy is the best response to the strategies chosen by the other players. No player can improve their payoff by changing their strategy alone while the other players keep their strategies unchanged.

Example: Prisoner's Dilemma

	Prisoner B: Cooperate	Prisoner B: Defect
Prisoner A: Cooperate	(-1, -1)	(-5, 0)
Prisoner A: Defect	(0, -5)	(-3, -3)

- If B cooperates, A gets a better outcome by defecting ($0 > -1$).
- If B defects, A still prefers defecting ($-3 > -5$).
- The same reasoning applies to B.

Nash Equilibrium: (Defect, Defect)

Importance in Game Theory

Nash Equilibrium is used in:

- Economics
- Business strategy
- Auctions
- Political science
- Negotiation and bargaining
- Marketing competition
- International relations

Field	Example	Nash Equilibrium
Economics	Two competing firms decide whether to charge a high price or a low price.	Both firms choose a low price because if one raises its price while the other keeps prices low, it loses customers.
Business Strategy	Coca-Cola and PepsiCo decide whether to launch major advertising campaigns.	Both advertise heavily because neither wants to lose market share if the other advertises.
Auctions	Companies bid for a government contract.	Each bidder submits a bid based on what they think others will bid; no bidder wants to change their bid after seeing competitors' strategies.
Political Science	Two political parties decide whether to campaign aggressively or moderately.	Both choose aggressive campaigning because neither wants to risk losing voters.
Negotiation & Bargaining	A labor union and a company negotiate wages.	They settle on a wage where neither side benefits from changing its position alone.

1. Mother and Vegetable Vendor (Bargaining Example)

Suppose a mother goes to a vegetable market.

- Vendor asks ₹60/kg for tomatoes.
- Mother offers ₹40/kg.
- Vendor refuses.
- Mother starts walking away.
- Vendor lowers to ₹50/kg.
- Mother increases to ₹48/kg.
- Finally they agree at ₹49/kg.

Nash Equilibrium

At ₹49/kg:

- If the mother lowers her offer further, the vendor may refuse to sell.
- If the vendor raises the price, the mother may buy from another shop.

Neither side can improve their outcome by changing their strategy alone.

Equilibrium = ₹49/kg deal.

This is a simple example of a **Bargaining Game**.

U.S.–Iran Conflict (Political / International Relations)

A simplified version of the recent situation:

Strategies

USA	Iran	
Escalate military action	Escalate military action	
Negotiate	Negotiate	
Possible Outcomes		
	Iran Escalates	Iran Negotiates
USA Escalates	War costs for both	USA gains leverage
USA Negotiates	Iran gains leverage	Peace agreement

During periods of tension:

- The U.S. does not want to appear weak.
- Iran does not want to appear weak.
- Both maintain military readiness.

This creates a **Security Dilemma**.

Nash Equilibrium (during conflict phase)

Often:

(Maintain pressure, Maintain pressure)

Neither side wants to unilaterally back down because the other side could gain a strategic advantage.

Nash Equilibrium (during negotiation phase)

If both believe continued war is too costly:

(Negotiate, Negotiate)

Game Theory Insight

The mother–vendor example is a **Bargaining Game**.

The U.S.–Iran example is closer to:

- **Prisoner's Dilemma**
- **Security Dilemma**
- **Chicken Game** (who backs down first?)
- **Repeated Games** (because the countries interact repeatedly over many years)